# Project Design Phase-II
# Data Flow Diagram & User Stories

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|---|---|
| Date | 31 January 2026 |
| Team ID | [Add Team ID] |
| Project Name | Global Fuel Prices & Taxation Analysis Dashboard |
| Maximum Marks | 4 Marks |

**Data Flow Diagrams:**

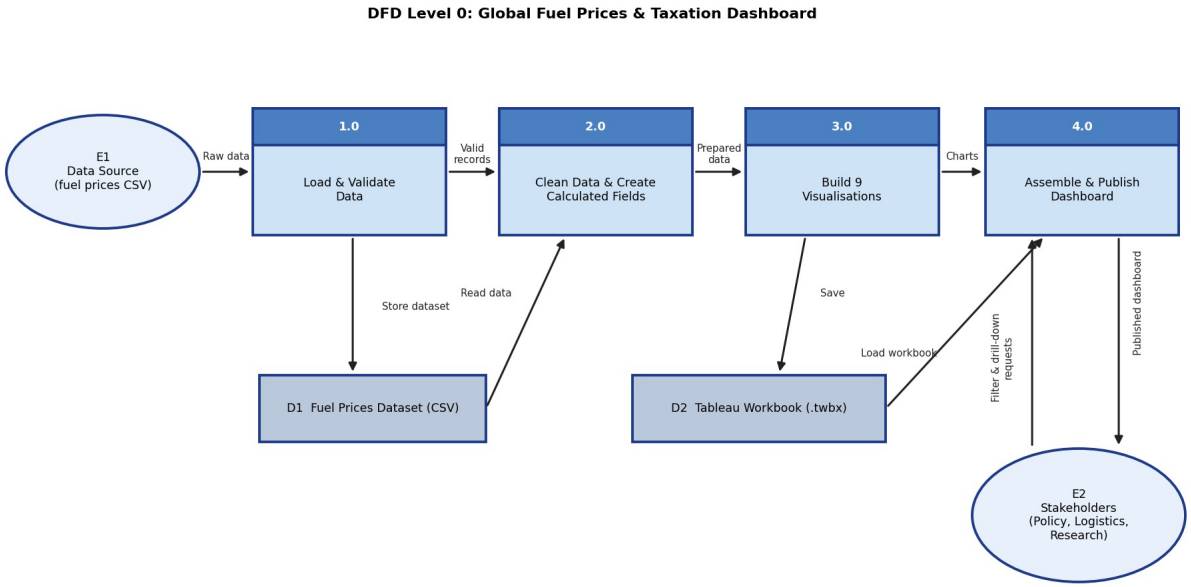
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

**DFD Symbol Legend**

| Symbol | Name | Description |
|---|---|---|
| **Oval / Rounded shape** | **External Entity** | A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway). |
| **Rectangle with numbered header** | **Process** | An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order). |
| **Rectangle (solid fill, no number)** | **Data Store** | A place where data is held for later use (e.g., Customer Database, Inventory). |

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|---|---|---|
| **Labeled arrow** | **Data Flow** | The movement of data between entities, processes, and data stores, labeled with what data is being passed. |

**DFD Level 0: Global Fuel Prices & Taxation Dashboard**



## User Stories

Use the below template to list all the user stories for the product.

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Policy Consultant | Data Access & Integration | USN-1 | As a policy consultant, I can load the global fuel dataset so that I can analyse tax and subsidy data for 84 countries. | Dataset loads with all 27,468 records and 10 fields | High | Sprint-1 |
| Policy Consultant | Tax & Subsidy Analysis | USN-2 | As a policy consultant, I can view a map of average fuel tax by country so that I can spot tax disparities. | Map shows tax rate per country and responds to the year filter | High | Sprint-1 |
| Policy Consultant | Tax & Subsidy Analysis | USN-3 | As a policy consultant, I can compare tax and subsidy levels by region. | Grid shows average tax for each region and subsidy level | High | Sprint-1 |
| Policy Consultant | Tax & Subsidy Analysis | USN-4 | As a policy consultant, I can see whether subsidies lower fuel prices. | Chart shows average fuel price for Low, Medium, High and Very High subsidy levels | High | Sprint-1 |
| Supply Chain Manager | Price Trends & Forecast | USN-5 | As a supply chain manager, I can see diesel price trends and a forecast by region so that I can plan procurement. | Line chart shows monthly diesel price per region with a 12-month forecast | High | Sprint-2 |
| Supply Chain Manager | Market Comparison | USN-6 | As a supply chain manager, I can see the 10 costliest and 10 cheapest diesel markets. | Ranked bars show the top and bottom 10 countries | Medium | Sprint-2 |
| Supply Chain Manager | Risk Analysis | USN-7 | As a supply chain manager, I can see diesel price volatility by region and year. | Heat map shows volatility with colour scale and | Medium | Sprint-2 |

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|---|---|---|---|---|---|---|
| | | | | labels | | |
| Energy Researcher | Income-Based Research | USN-8 | As a researcher, I can compare average fuel tax across income groups. | Bars show average tax for Low, Middle and High income groups | High | Sprint-2 |
| Energy Researcher | Income-Based Research | USN-9 | As a researcher, I can see how subsidy levels are spread across income groups. | 100% stacked bars show subsidy level share per income group | Medium | Sprint-3 |
| Energy Researcher | Income-Based Research | USN-10 | As a researcher, I can compare petrol, diesel and LPG prices by income group. | Table shows average price per fuel type and income group | Medium | Sprint-3 |
| All users | Dashboard Interaction | USN-11 | As a user, I can filter by year and region and click a chart to filter the others. | All charts update when a filter or selection changes | High | Sprint-3 |
| Administrator (Analyst) | Publishing | USN-12 | As the analyst, I can publish the dashboard to Tableau Public and share the link. | The public link opens the full dashboard | Medium | Sprint-3 |